

1- Carson Inc.'s manager believes that economic conditions during the next year will be strong, normal, or weak, and she thinks that the firm's returns will have the probability distribution shown below. What's the standard deviation of the estimated returns?

Conditions	Prob.	Return
Strong	30%	30.0%
Normal	40%	11.0%
Weak	30%	-18.0%

2- Returns for the Dayton Company over the last 3 years are shown below. What's the standard deviation of the firm's returns?

<u>Year</u>	<u>Return</u>
1	18.00%
2	-12.50%
3	27.50%

3- Company A has a beta of 0.60, while Company B's beta is 1.10. The required return on the stock market is 9.00%, and the risk-free rate is 2.25%. What is the difference between A's and B's required rates of return?

4- Mikkelson Corporation's stock had a required return of 8.65% last year, when the risk-free rate was 2% and the market risk premium was 4.75%. Then an increase in investor risk aversion caused the market risk premium to rise by 2 percentage points. The risk-free rate and the firm's beta remain unchanged. What is the company's new required rate of return?

5- Rosenfeld Corp believes the following probability distribution exists for its stock. What is the coefficient of variation on the company's stock?

State of the Economy	Probability of State Occurring	Stock's Expected Return
Boom	0.25	25%
Normal	0.55	15%
Recession	0.20	5%

6- Jack holds a \$100,000 portfolio consisting of the following stocks. What is the portfolio's expected return?

Stock	Investment	Expected return
A	\$10,000	25%
B	\$30,000	10%
C	\$20,000	5%
D	\$40,000	30%
Total	\$100,000	

7- Omar Angel holds a \$200,000 portfolio consisting of the following stocks. What is the portfolio's beta?

Stock	Investment	Beta
A	\$50,000	0.60
B	\$50,000	1.60
C	\$50,000	1.00
D	\$50,000	0.50
Total	\$200,000	

8- Assume that you are the portfolio manager of the SF Fund, a \$6 million hedge fund that contains the following stocks. The required rate of return on the market is 10.50% and the risk-free rate is 1.70%. What rate of return should investors expect (and require) on this fund?

Stock	Amount	Beta
A	\$1,680,000	1.20
B	\$1,470,000	0.50
C	\$1,950,000	1.40
D	\$900,000	0.75
Total	\$6,000,000	

9- You hold a diversified \$100,000 portfolio consisting of 10 stocks with \$10,000 invested in each. The portfolio's beta is 1.18. You plan to sell a stock with $b = 0.70$ and use the proceeds to buy a new stock with $b = 1.70$. What will the portfolio's new beta be?

